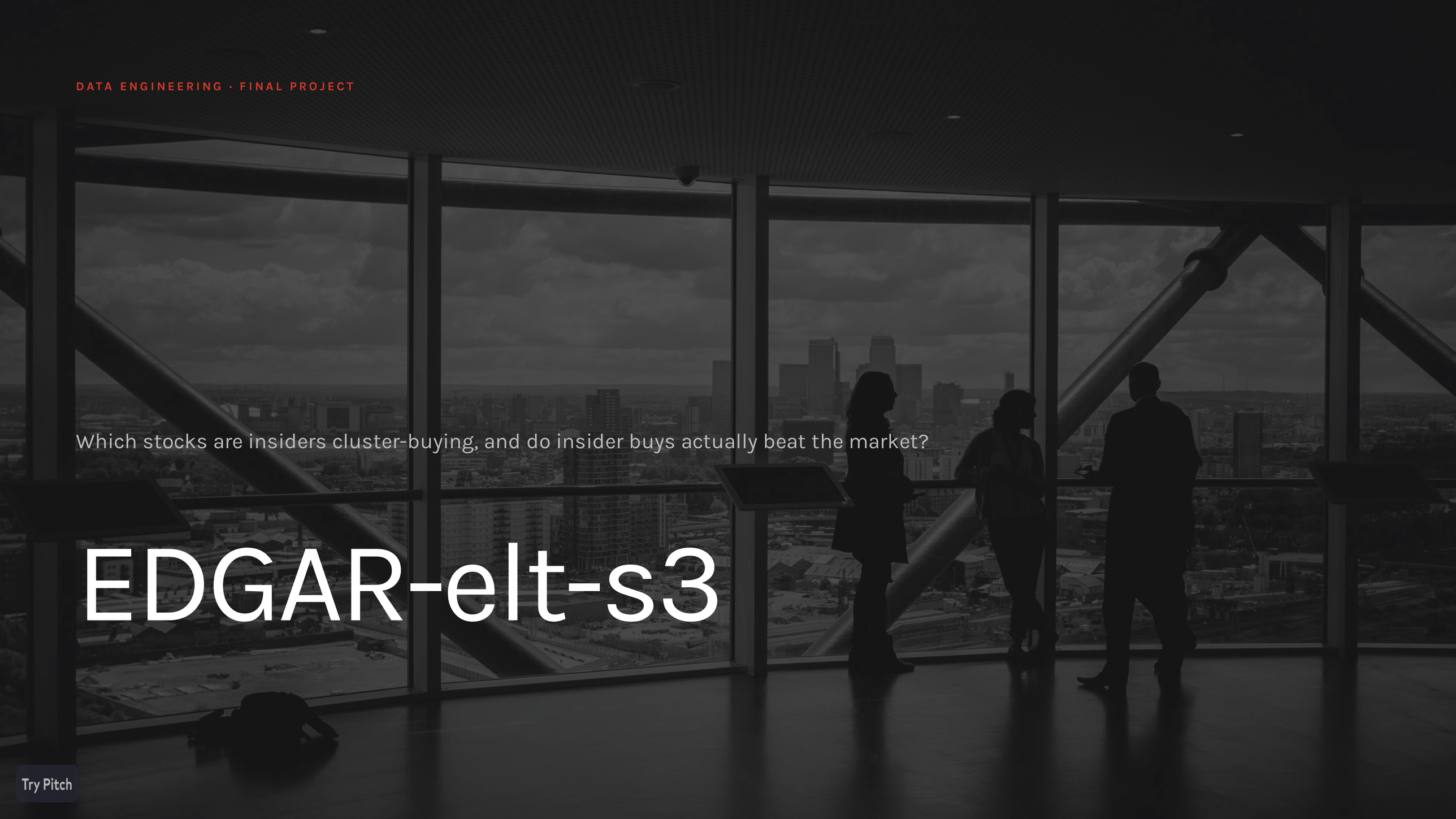
DATA ENGINEERING · FINAL PROJECT



Which stocks are insiders cluster-buying, and do insider buys actually beat the market?

EDGAR-elt-s3

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01 · THE DATA

SEC EDGAR

What it is

Insider trades (Forms 3, 4, 5), financial statements (10-K, 10-Q), CIK ↔ ticker list, and daily stock prices.

Where it came from

SEC insider transaction and financial statement data sets, SEC company_tickers_exchange.json, and Yahoo Finance via yfinance.

License and update frequency

SEC data is public domain, accessed per SEC fair access policy (declared User-Agent, max 10 req/s). Filings update quarterly; tickers and prices daily.

Why I picked it

Insiders must report every trade within two business days. The data is free, official, and goes back 20 years, and it answers a real investing question.

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What I set out to answer

“By the end of this, I'll be able to answer who is cluster-buying and whether it pays, about insider trading in US stocks.”

QUESTION 01

Which stocks are insiders cluster-buying right now?

QUESTION 02

Do insider buys actually beat the market?

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Five choices, wired end to end

① SOURCE

SEC EDGAR + yfinance

Forms 3/4/5, 10-K/10-Q, tickers, daily prices

② RAW LANDING

RustFS (S3)

Raw files saved untouched, replayable

③ ANALYTICAL STORE

ClickHouse

Bronze → silver → gold, plus dbt data tests

⑤ SERVING

Streamlit

Dashboard answering both questions from gold

④ TRANSFORM + ORCHESTRATION

dbt + Prefect · extract → land → load → dbt build (22+ tests) · weekdays 6am Central · retries on every network step

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What the data says

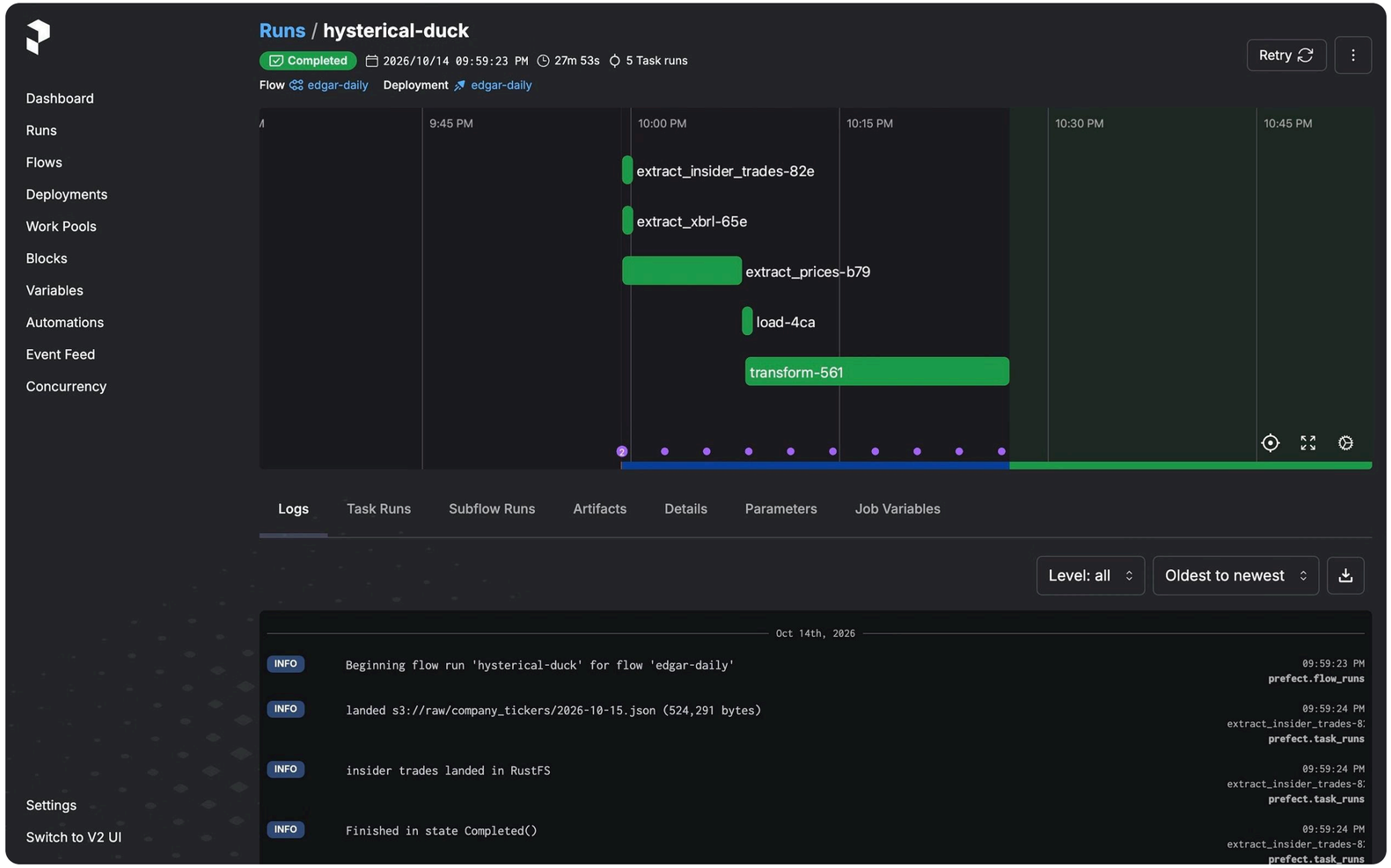


STREAMLIT DASHBOARD

- **Mostly no.** Across 123,928 insider buys, the median 180-day edge over SPY is **–1.4 pts**
- Only **47%** of buys beat SPY over the same days
- Insiders beat SPY in a few years (2016, 2020) but trailed it in most others
- View 1 (cluster buys) lists the stocks several insiders are buying right now

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Prefect run history



SCHEDULED DEPLOYMENT

- **Run “hysterical-duck”: Completed** on Oct 14, 2026 in 27m 53s, 5 task runs
- Flow: extract (insider trades, XBRL, prices) → load → transform
- Raw files land in RustFS, keyed by date (e.g. company_tickers/2026-10-15.json)
- Cron: every weekday, 6am Central
- Retries on every network step (SEC, Yahoo)
- Runs in Docker, with no open terminal needed

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What broke and what I'd do next

WHAT BROKE

- **SEC moved its files.** From 2026q2 the insider zips have a new URL. The 404s were treated as “not published yet,” so two quarters were silently skipped. The extractor now uses the new path.
- **SEC's own data has gaps.** 2013q1 has 1,029 XBRL numbers for filings missing from its sub.txt. The relationships test now flags these as a warning.
- **Memory.** Sorting to dedupe 27M prices ran ClickHouse out of memory. A row_number() window does it in seconds.
- **Dates before 1970** collapsed silently until silver switched to Date32.

WHAT I'D DO NEXT

- **Close the lag:** add a daily feed from EDGAR's filing index (~1,000 XML filings/day) instead of quarterly bulk files
- **Alert on failures:** notify when a fetch 404s or returns nothing, so no quarter is ever skipped silently
- **Go to the cloud:** RustFS is S3-compatible, so the same code can run on AWS S3
- **Sharper benchmark:** compare insider buys against sector peers, not only SPY

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